**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 031 · Network Attacks

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. How does phishing exploit users?

Q2. A star network's central switch fails. What happens?
A. Only one cable fails
B. All files are deleted
C. Connected devices lose contact
D. It becomes a mesh

Q3. Why can a mesh topology be resilient?

Q4. (a) Convert binary 10110110 to hexadecimal, using four-bit groups. (b) Convert hexadecimal B6 into eight-bit binary.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

Brute-force attack — Repeated guessing of possible passwords.

DoS — Denial of Service; overloads availability of a service.

DDoS — Distributed Denial of Service; traffic from multiple sources.

Data interception — Capturing information while it is transmitted.

SQL injection — Using unsafe input handling to affect database queries.

Unauthorised access — Access without proper permission.

## New learning

Brute-force guessing: An attacker systematically tries many possible passwords in an attempt to gain access to an account.

Denial of Service: A DoS attack overwhelms a service with traffic or requests so legitimate users cannot access it. A DDoS attack originates from multiple systems.

Data interception: Data may be captured during transmission, potentially exposing confidential information if not protected.

SQL injection: A malicious input may be interpreted as part of a database query, causing unauthorised viewing or alteration of data if the application handles input insecurely.

## Worked example

A school's web service receives overwhelming requests and genuine users cannot log in. Explain the likely attack and its effect. [4 marks]

1. The service is overloaded and genuine users cannot access it.

2. A denial-of-service (DoS) attack targets service availability [1].

3. Excess traffic or requests [1] overload server resources [1].

4. Legitimate pupils and staff are prevented from accessing the website [1].

# We Do · Practise with guidance

### Guided 1

An account receives thousands of unsuccessful login attempts. Explain the attack and its likely aim. [4 marks]

Worked method: A brute-force attack [1] tries many possible passwords [1] seeking the correct combination [1] to gain unauthorised access [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## Guided 2

A user sends private details on an unprotected connection. Explain data interception and a consequence. [4 marks]

Worked method: Data interception captures data during transmission [1] without authorisation [1]. Private details may be read [1] and misused for impersonation or theft [1].

## Guided 3

An insecure database-backed website misinterprets crafted input as part of a query. Explain the threat and possible damage. [4 marks]

Worked method: SQL injection occurs when unsafe user input changes a database query [1] so unintended commands are executed [1]. This may reveal private records [1] or alter/delete stored data [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Code First · Predict → Write → Test → Improve

Write the final line that shows the number of failed attempts.

```
failed = 2
failed = failed + 1
```

write

### Question 1

Define a brute-force attack. [2 marks] Source: Adapted from your Paper 1 revision booklet, 1.4.1.

### Question 2

A museum's online ticket-booking service is deliberately overwhelmed with requests. Explain TWO consequences of the denial-of-service attack for the museum and its visitors. [4 marks] Source: Original OCR J277-style question, not a past-paper original.

### Question 3

Describe DoS, interception and SQL injection, including an effect of each. [6 marks] Source: Original OCR J277-style question, not a past-paper original.

# Exit Ticket · Answer before checking

| Do Now | I Do | We Do | You Do | Review & Improve |
|---|---|---|---|---|
| Recall previous learning | Read new knowledge | Practise with guidance | Work independently | Check understanding |

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

A DoS attack aims to make a service unavailable to legitimate users.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

## Page 1 onwards · Do Now

Q1. Phishing pretends to be a trusted source to trick people into sharing data or doing something unsafe.

Q2. C — Connected devices lose contact

Q3. Other paths can keep data moving if some links fail. More links add cost and complexity.

Q4. (a) B6: 1011 0110. (b) 10110110.

## I Do / We Do pages · Guided practice

Q1. A brute-force attack [1] tries many possible passwords [1] seeking the correct combination [1] to gain unauthorised access [1].

Q2. Data interception captures data during transmission [1] without authorisation [1]. Private details may be read [1] and misused for impersonation or theft [1].

Q3. SQL injection occurs when unsafe user input changes a database query [1] so unintended commands are executed [1]. This may reveal private records [1] or alter/delete stored data [1].

## You Do pages · Code First

```
failed = 2
failed = failed + 1
print(failed)
```

Expected result: 3

## You Do pages · Independent questions

Q1. Attempts many possible passwords [1] to discover the correct one/gain access [1].

Q2. The booking website becomes overloaded or unavailable [1], so genuine visitors cannot buy or view tickets [1]. The museum may lose bookings/revenue or have increased support work [1], disrupting museum operations [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Q3. DoS overloads a service [1], causing legitimate access failure [1]. Interception captures data in transit [1], risking disclosure [1]. SQL injection manipulates database queries via unsafe input [1], potentially revealing or changing stored records [1].

### Exit Ticket · Model answer

True

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**